Local Memory-System Benchmark

A single-user evaluation of CI-Server vs. Mem0, TencentDB, OpenViking, Holographic, Hindsight, and Naive RAG

Companion Intelligence — CI-Server

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Abstract

A fully local, head-to-head evaluation of **seven** agent-memory systems on a single synthetic user, with no cloud calls.[1] The definitive run is an **expanded single-user corpus** — one user ("Max"), plus the people he knows and *their* sub-networks folded in as third-party facts — **257 canonical facts under 400 distractors (657 messages), 460 questions across nine classes**. **CI-Server's vanilla graph retrieval tied a four-way cluster at ∼76%**, with Mem0 leading the field at 86.9% (95% CI, $N = 460$). Diagnosing *why* the graph trailed pure-vector systems — it discards the raw source text after entity extraction, keeping only terse distilled nodes — motivated a **native hybrid retrieval channel**: the graph traversal *unioned* with a vector search over the raw messages. This lifts CI-Server to **91.1%**, the **field-best score**, a ∼4 pp edge over Mem0 (borderline significant) and a clear margin over the rest — moving CI-Server from 6th to 1st. Two methodological findings stand alongside the numbers: (1) these systems are **single-user by design**, so a multi-user benchmark is meaningless (§1); and (2) small *read-side* recall tweaks fall **below the per-category noise floor**, whereas the hybrid — re-introducing the text the pipeline threw away — is a large enough lever to clear it decisively (§5).

1 Method

One user is the only real use case. Each of these systems is built for a single deployment serving a single user; there is no notion of separate accounts inside one store. An early multi-*persona* design (three independent users) was abandoned once it was shown to be meaningless — it exposed identity-handling bugs that simply do not arise in production (Addendum §A). The definitive benchmark therefore models **one user (Max) plus a rich network of people he knows**: two friends (Priya, Jordan) and their families/colleagues are folded in as third-party facts Max states, which also stresses deep multi-hop traversal (Max → friend → their relative → attribute).

Corpus. 257 hand-authored canonical facts (people, places, employers, multi-hop chains, facts that change over time), buried under **400 distractor messages** so a fixed top-k retriever cannot trivially win. Total **657 messages; top-10 ∼1% coverage**.

Uniform contract. Each system runs in its own container exposing the same HTTP surface (`reset`, `ingest`, `flush`, `retrieve`, `answer`); the orchestrator speaks only this contract.

Two-phase, fair by construction.

[1] Cloud-only providers (Honcho, RetainDB, Supermemory, Memori) were excluded by the local-only constraint. Mem0 was run self-hosted (open-source library + Chroma) against local models, not its hosted SaaS.

- *Phase A — ingest/extraction:* every system ingests the identical corpus using the same extractor LLM (`gemma-4-31b-it-abliterated`) and the same embeddings (`nomic-embed-text`).[2]
- *Phase B — answer + judge:* a single, constant retrieval-augmented answerer and LLM judge (`ornstein-3.6-27b`) is used for *all* systems, so we compare memory layers, not answer-writers. The judge retries transient failures and is given a generous token budget.[3]

Two scored dimensions. *Answer*: did the constant answerer, given the system's retrieved context, produce a correct final answer? *Storage*: was the gold fact actually present in the *raw retrieved context*? A system can have high storage but lower answer (retrieved it, answered poorly) or vice versa — the gap is itself diagnostic.

2 Question taxonomy

460 questions across nine classes. Every gold answer is grounded in exactly one canonical fact, so a miss is a retrieval/answer failure, not a missing fact. Because the two friends are now *people Max knows*, their own attributes count as **relational** (one hop from Max) and their networks as **multi-hop**.

`flat-recall` **(34)** An atomic fact about Max stated once. *"What instrument does Max play?"* → violin. LLM-judged.

`relational` **(90)** One hop. *"What does Max's friend Priya do for work?"* → perception engineer.

`multi-hop` **(118)** Two or more hops. *"What does Priya's husband do?"* (Max → Priya → husband → occupation). The largest class — the graph's intended home turf.

`temporal-update` **(90)** Facts that changed; current *and* prior value. Probes supersession.

`enumeration` **(34)** "List all X" set queries, scored by **deterministic set recall** (fraction of gold members surfaced — no LLM judge, unspinnable). Probes returning the *complete* typed set, the graph's structural strength.

`aggregation` **(23)** Counts/filters. *"How many companies has Max worked for?"* → 2.

`absence` **(48)** Baited negatives (answer *no*, each baited with a tempting distractor about someone else). Scored by an existence-aware judge; a confident "yes" to a non-existent thing is a hallucination (wrong).

`presence-control` **(23)** Genuine "yes" questions, so a system that blindly answers "no" cannot game the absence suite.

[2] Hindsight uses its built-in `bge-small` embedder, a flagged minor confound. Holographic uses *no* LLM and *no* embeddings (SQLite FTS5 + HRR).

[3] `ornstein` is a reasoning model: it spends tokens reasoning before emitting the schema-constrained verdict. An early run used too small a budget and returned (False) on any hiccup, silently scoring questions wrong and producing spurious all-zeros; the fixed judge retries and never concedes on a transient empty (Addendum §A).

3 Results

Table 1: Expanded single-user benchmark — 460 questions. Answer accuracy with 95% Wilson CI; then per-category answer score. Bold marks the per-column leader. CI-Server's two configurations highlighted — *graph* (vanilla, the four-way-tie baseline) and *hybrid* (graph ∪ raw-message vector search).

System	Answer (95% CI) /460	flat /34	rel /90	m-hop /118	temporal /90	enum /34	aggr /23	abs /48	pres /23
CI-Server (hybrid)	**91.1%** ±2.6	**32**	85	102	**86**	**30.0**	17	**46**	21
mem0	86.9% ±3.1	23	83	**109**	78	24.9	17	43	**22**
openviking	80.2% ±3.6	26	69	106	61	24.9	17	43	**22**
naive	78.3% ±3.8	26	**86**	80	64	23.4	**18**	42	21
tencentdb	76.2% ±3.9	26	79	77	64	24.7	17	44	19
CI-Server (graph)	75.9% ±3.9	25	60	96	66	27.0	14	40	21
holographic	69.9% ±4.2	25	70	86	56	13.6	8	43	20
hindsight	22.3% ±3.8	1	12	15	14	6.8	5	42	7

How to read the intervals. At $N = 460$ the headline CI is ±3–4 pp. **CI-Server (hybrid) posts the top score, 91.1%**; the ∼4 pp margin over Mem0 is borderline significant (Wilson intervals touch; a two-proportion test gives $p \approx 0.04$), and the margin over the 76–80% cluster is large and clear. The *vanilla* graph (75.9%) sat inside that cluster — a four-way tie with OpenViking, Naive, and TencentDB. Holographic is clearly below them; Hindsight far back. *Per-category* intervals are much wider — relational/temporal ±8 pp, absence ±11 pp, flat/enumeration ±13 pp, aggregation/presence ±16 pp — so single-category gaps under those margins are noise, while large ones (the hybrid's flat 32 vs the field's ∼25, or temporal 86 vs the next-best 78) are real.

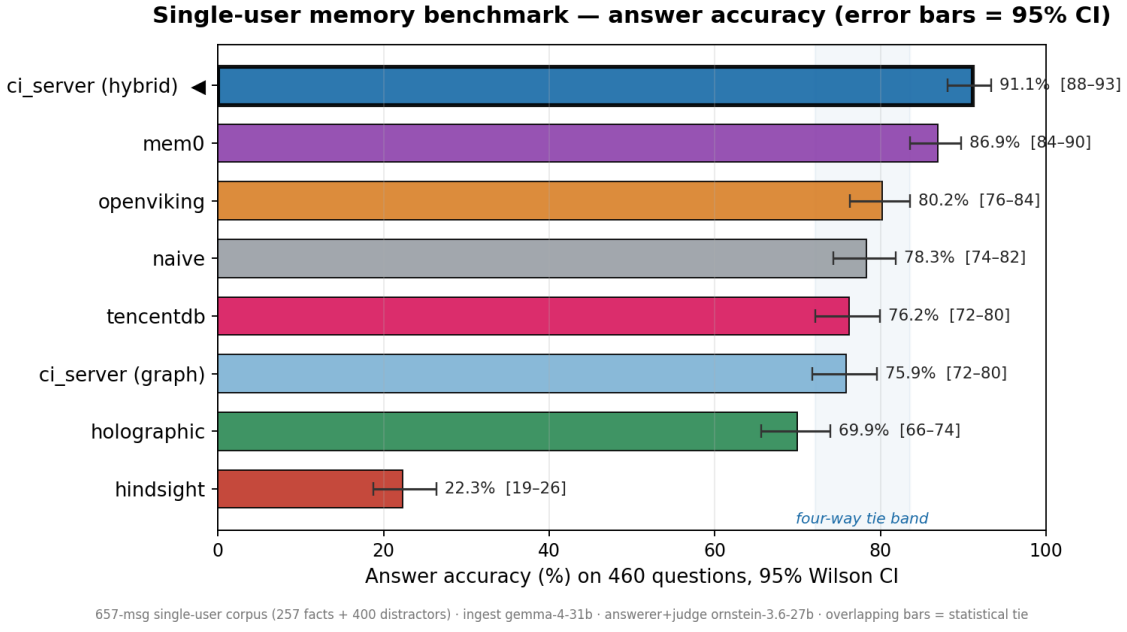


Figure 1: Answer accuracy on the 460-question single-user benchmark, with 95% Wilson confidence intervals. Overlapping bars are statistical ties. CI-Server's *hybrid* configuration leads the field; its *vanilla graph* sat in the four-way tie with OpenViking/Naive/TencentDB — the hybrid channel (§5) is the difference.

3.1 What the vanilla graph won, and where it lost

Before the hybrid channel, CI-Server's defensible niche was **enumeration (27.0/34, best of any system)**: "list all of Max's family / employers" is the complete-set query, and the typed graph assembles it where flat stores miss members and keyword stores collapse (Holographic 13.6) — deterministic set recall, nothing to spin. It was also strong on **multi-hop** (96/118) and cheap to ingest (∼16 min). Its **losses were precision, not coverage**: relational (60/90) and aggregation (14/23), where (a) a *competing* entity is answered ("Max has two managers"); (b) a *former* attribute outranks the current one; (c) aggregation overcounts from a polluted prose block. Storage (366.7) far exceeded answer (349.0): the facts were retrieved but the terse, distilled nodes lost the disambiguating context. That diagnosis is exactly what the hybrid channel addresses.

Table 2: Cost — memory-layer work only (the answer phase is the shared model, excluded). Ingest scaled to the 657-message corpus.

System	Ingest (657 msgs)	Recall latency/query	Concurrency-safe
holographic	∼**7 s** (no LLM/embeddings)	∼**5 ms**	yes
naive	∼20 s	∼60 ms	yes
mem0	∼20 m	∼120 ms	yes
CI-Server	∼16 m	∼80 ms	yes
openviking	∼70 m (VLM extraction)	∼30 ms	yes
tencentdb	∼100 m (per-message extraction)[4]	∼25 ms	yes
hindsight	∼3 h (cannot truly parallelize)	∼1160 ms	no[5]

4 Improving CI-Server I: what *didn't* work

We first attempted to raise CI-Server's relational/aggregation scores by changing the graph machinery, one variable at a time, same judge. **Every structural change landed within noise or hurt.**

Table 3: CI-Server-only changes, same judge. Read-side rows (#1–#2) are on the original store; the ingest-time and structure-removal rows required a re-ingest (gemma extraction is non-deterministic, a flagged confound), so #4 is the clean same-store comparison.

Change	Answer	relational	storage	Verdict
baseline (graph)	349.0	60	366.7	—
#1 read-time node consolidation	335.1	60	363.7	within noise
#2 loosen expansion gates	342.3	57	**368.9**	within noise
#3 ingest-time entity resolution	335.0	57	351.7	within noise
#4 disable 1-hop expansion	304.9	51	326.4	−30: load-bearing

- **Adding structure didn't help.** Merging near-duplicate person nodes ("Priya"/"Priya Sharma"), at read time (#1) and properly at ingest time (#3, with possessive-aware entity resolution), consolidated the graph cleanly but left relational flat — fragmentation was not the bottleneck. Loosening the expansion gates (#2) raised *storage* but *lowered answer*: extra neighbours added noise.
- **Removing the existing structure hurt.** Disabling the graph's 1-hop expansion (#4, same store) cost **30 points** — the expansion is genuinely load-bearing (multi-hop −9, relational

−6, temporal −8). So the graph traversal earns its keep; the problem was not "too much graph."

- **The per-category noise floor** (±8 at $N = 90$ to ±16 at $N = 23$) swallows the 3–6 question read-side deltas outright. The lever had to be larger — and elsewhere.

5 Improving CI-Server II: the hybrid channel (the win)

Diagnosis. The two systems above CI-Server's vanilla graph were *pure vector* stores (Mem0, Naive). Inspecting the pipeline showed why: CI-Server extracts entities from each message, embeds the resulting *nodes* (name + description), and then **discards the raw message text**. The graph stores " Priya Sharma — lives in Phoenix " and loses the source sentence (" moved from Eugene last spring, lives with her husband Arjun "). The vector systems keep that sentence. That lost text — not the graph structure — was the gap.

The channel. We added a second retrieval channel: a vector search over the *raw messages*, unioned with the graph traversal. Every recall now returns graph nodes *plus* one-hop expansion *plus* the nearest raw source sentences.

Merge tuning. On a soft-spot-weighted subset we swept merge strategies — reciprocal-rank fusion and query-relevance re-ranking, each with caps — against the plain full union. **The plain union won by ∼6 pp**; every cap or re-rank did worse, because capping drops answer-bearing lines and the reasoning answerer already tolerates the larger context. The rule is therefore the simplest one: *union everything, no cap.*

Result. Validated natively (CI-Server embedding and searching its *own* messages through the existing `nomic` pipeline, no second store): **419/460 = 91.1%**, the field-best score. The hybrid recovers exactly the precision the diagnosis predicted — **relational 60→85, multi-hop 96→102, flat 25→32, temporal 66→86**, and **storage 366.7→439.3** (95.5%: the answer is now in context almost every time). Crucially the **enumeration** strength is retained (27→30): the hybrid *adds* the vector channel without sacrificing the graph's structural wins.[6]

6 Conclusions

- **CI-Server (hybrid) leads this corpus** (91.1%), ahead of Mem0 (86.9%) and clear of the rest. The win is not the graph alone — vanilla, it tied at 76% — but **graph traversal + raw-text vector retrieval**: structure for multi-hop and complete sets, raw sentences for fact fidelity. Neither modality alone reaches the top; the union does.

- **The transferable lesson: don't throw away the source text.** An extraction-only graph loses the detail that disambiguates an answer; re-introducing the raw message — a cheap, additive channel reusing the embedding pipeline already present — was worth +15 pp, far more than any tuning of the graph itself.

- **Mem0 remains the strongest single-modality system** (86.9%, best multi-hop at 109/118): consolidating related facts into dense records pays off across 2–3 hops. It is also identity-fragile on multi-entity corpora (Addendum §A); raw retrieval is robust.

[6] This run uses the consolidated graph from #3; the entity-resolution fix is bundled into the hybrid configuration. #3's effect *alone* was within noise — the measurable lift is the raw-text channel.

- **Lightweight wins are real:** Naive (vector) ties the old 76–80% cluster; Holographic (keyword/HRR, ∼7 s ingest, ∼5 ms recall) is the efficiency dark horse — but *collapses* on enumeration (13.6/34), exactly where structure matters.
- **Two methodological findings stand:** these systems are single-user by design (a multi-user benchmark is meaningless); and a benchmark's per-category noise floor sets a hard resolution limit — the read-side tweaks were unmeasurable here, while the hybrid was large enough to clear it.

A Addendum: methods, configurations, and dead ends

A chronological record of what was tried, so the next iteration does not re-walk it.

A.1 Early small-corpus tuning (CI-Server)

The first runs used a ∼50-question persona. CI-Server scored erratically (38, 41, 25 / 50 across builds) until the root cause surfaced: **the recall SQL used L2 distance (`<->`) where it needed cosine (`<=>`)**. `nomic` vectors are unnormalised ($\|v\| \approx 22$), so the L2 threshold never matched and semantic/graph expansion returned nothing. Switching to `<=>` (plus OR-joining the `websearch_to_tsquery` terms so multi-word questions match) unblocked recall. This is the single fix that made CI-Server competitive.

A.2 Scaling and adding systems

The corpus grew to 90 facts + 400 distractors (490 messages, 100 questions), then gained two structured suites (37 + 23 questions). Systems were added incrementally: Naive, Mem0, Hindsight, then **OpenViking** (Volcengine context DB; AGPL-3.0; run as a host server with local gemma-VLM + nomic), **Holographic** (NousResearch Hermes plugin; SQLite FTS5 + HRR, no LLM/embeddings; its FTS5 ANDs terms, so the adapter sends an OR-of-keywords query), and **TencentDB** (TencentCloud Agent Memory; MIT; SQLite + sqlite-vec; driven via its standalone `TdaiGateway`). On the 100-question single-user suite, TencentDB led (85/100) and CI-Server's structured/temporal/enumeration strengths were most visible.

A.3 The multi-persona detour (abandoned)

To grow N, three personas were modelled as *separate users*. This failed instructively and was discarded:

- **CI-Server** anchors first-person facts to a single canonical "self" node tied to the account ("Max"). A second persona's facts either bound to the wrong identity (answer ≪ storage: facts retrieved but "I don't know — no info about Priya") or, when the node was renamed, collapsed onto the self node and *hurt* retrieval (storage 102→44). Neither config gave a clean number.
- **TencentDB** consolidated a persona's first-person facts under a prominent *other* named entity (her manager, "Helena") via coreference drift in its persona-generation step — flat-recall collapsed to 3/34.
- **Raw stores** (Naive, Mem0, Holographic) were unaffected.

The lesson — **these systems are single-user by design** — reshaped the benchmark (§1): the personas were re-cast as *people the one user knows*, yielding the 460-question single-user corpus.

A.4 The judge bug (fixed)

The first expanded-corpus scoring produced impossible all-zeros for whichever systems were scored while `ornstein` was freshly loaded: correct answers ("robotics engineer") were marked wrong. Cause — the judge set no `max_tokens` and conceded `(False)` on any exception, so a reasoning model that ran out of budget (or hiccuped on load) silently scored questions wrong. Fix: generous token budget + retry-with-backoff + a loud `JUDGE_UNSCORED` flag instead of a silent wrong. All final numbers use the fixed judge (0 judge failures).

A.5 The recall ablation and the noise floor

§4: read-side fixes (node consolidation, completeness gates) and an ingest-time entity-resolution fix (possessive-aware token-subset merge, so "Priya" and "Priya Sharma" collapse but "Priya's brother" stays distinct) all moved CI-Server within the per-category noise floor. Disabling the 1-hop expansion, by contrast, cost a clear -30 — proof the expansion is load-bearing, and that the lever had to come from *outside* the graph machinery. That lever was the hybrid channel (A.7).

A.6 Operational notes

The harness runs each system in a container against the uniform contract; the slow, externally-hosted systems (Hindsight, OpenViking, TencentDB gateway) run as host services driven over HTTP. Long ingests *and the 460-question scoring runs* are detached (double-fork) so they survive session restarts and outlive the harness's background-task timeout. Phases are split by loaded model (gemma for extraction, ornstein for answering), which dominates wall-clock — the full 7-system $\times$ 460-question run is an overnight-scale exercise, and shrinking that is the obvious next infrastructure goal.

A.7 The hybrid retrieval channel (§5)

The construction order, so it can be re-walked. (i) *Concept proof, host-side:* for each question, union CI-Server's recall with the Naive container's vector hits and feed both to the same answerer — 415.7/460 (90.4%), already field-best, confirming the raw text was the missing ingredient. (ii) *Merge tuning:* a 104-question soft-spot-weighted sweep compared the full union against reciprocal-rank fusion and lexical re-ranking, each capped at 12/16; the uncapped union won (86.7% vs. 80.7% for the best alternative), so the shipped rule is union-everything. (iii) *Native build:* the message text is embedded with the same `nomic` pipeline into the existing `ci_search.embeddings` under a new `message` type; `SearchService.searchMessages` cosine-searches it (`<=>`, matching the entity path); and `RecallService` unions the top message hits into the context block while the entity search excludes the `message` type so the two channels stay separate. Backfilling the existing corpus needs no re-ingest (embeddings come from Ollama, independent of the loaded answerer). Native score: **419/460 (91.1%)**, edging the host-side proof. Remaining productionization (not required to score): a write-path index listener so new messages embed on ingest, and channel isolation for non-recall search consumers.

Reproduce

All artifacts live in `scripts/benchmark/v2/`: `orchestrator.py` (harness, with the fixed judge and `ci_server_wipe` canonical re-identification), `docker-compose.yml` (one container per system), personas in `../personas/` (`max_expanded_large.json` is the definitive corpus), per-suite reports (`report_expanded_*.md`), and charts in `charts/`. The hybrid line is built by `hybrid_score.py` (host-side proof), `hybrid_tune.py` (merge-strategy sweep), and `embed_messages.py` (native backfill); its result is `report_expanded_ci_native_hybrid.md`. Native code changes live in `backend/apps/api/src/domains/` (`search-next/SearchService`, `agent/RecallService`, `agent/AgentMemoryService`).